

Symplectic Matrices in Control

Amp

A symplectic matrix preserves a bilinear form on phase space, and that single algebraic condition turns out to rigidify a spectrum, to explain why an undamped mechanical system cannot settle, and to sit underneath the Riccati equation that every LQR gain comes from. This note asks six questions from zero: what the symplectic condition preserves and forces about the inverse and determinant; why the flow of a Hamiltonian system is symplectic while a damped one is not; what the spectrum of a symplectic matrix must look like and why that rules out asymptotic stability; where a Hamiltonian matrix enters optimal control and how its invariant subspace produces the stabilizing Riccati solution; what a numerical reduction must preserve to compute that subspace honestly; and what changes when a pair of eigenvalues reaches the imaginary axis. The running example is a unit mass on a unit spring, first in phase space and then under an LQR cost. Answers are deliberately withheld.

October 01, 2026

Contents

1. What does the symplectic condition preserve?	3
2. Why is the flow of a Hamiltonian system symplectic?	4
3. What does symplecticity force on the spectrum?	5
4. Where does a Hamiltonian matrix enter control?	6
5. What must the numerical reduction preserve?	7
6. What happens when an eigenvalue lands on the imaginary axis?	8
7. Sources and further reading	9

1. What does the symplectic condition preserve?

Question: The condition $M^T J M = J$ is a statement about a bilinear form on $\mathbb{R}^{2n}$. What does that form measure, what does the condition force about M^{-1} and about $\det M$, and what does it demand in $2n$ dimensions beyond keeping volume fixed?

Running example. Write a phase-space vector as $x = \begin{pmatrix} q \\ p \end{pmatrix}$ with $q, p \in \mathbb{R}^n$, and let

$$J = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}, \quad \omega(u, v) = u^T J v. \quad (1)$$

A real $2n \times 2n$ matrix M is **symplectic** when

$$M^T J M = J. \quad (2)$$

The form ω is antisymmetric, $\omega(u, v) = -\omega(v, u)$. For the figures below take $n = 1$, so $x = \begin{pmatrix} q \\ p \end{pmatrix}$ is a point of the plane and $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$.

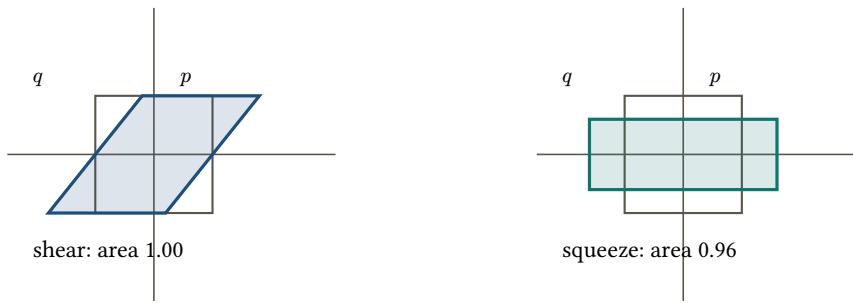


Figure 1: The unit square in the (q, p) plane and its image under two linear maps. Inspect what each image does to area, and how the shear image differs from the squeeze image.

Facts to reason with.

- For $n = 1$ the identity $M^T J M = (\det M) J$ holds for every 2×2 matrix M .
- The shear $S = \begin{pmatrix} 1 & 0.8 \\ 0 & 1 \end{pmatrix}$ has $\det S = 1$ and $\|S^T J S - J\| = 0$.
- The rotation by 30° has $\det = 1$ and $\|M^T J M - J\| = 0$.
- The squeeze $D = \begin{pmatrix} 1.6 & 0 \\ 0 & 0.6 \end{pmatrix}$ has $\det D = 0.96$ and $\|D^T J D - J\| = 0.04$.
- J itself satisfies $J^T = -J$, $J^2 = -I$, and $J^{-1} = -J$.

2. Why is the flow of a Hamiltonian system symplectic?

Question: For $\dot{x} = J\nabla H(x)$ the flow map Φ_t is symplectic for every t . Why must that be so, and which structural feature of the right-hand side is responsible? What does adding damping change, and what stays true of the undamped flow no matter how long it runs?

Running example. A unit mass on a unit spring has energy

$$H(q, p) = \frac{q^2 + p^2}{2}, \quad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \dot{x} = J\nabla H(x) = \begin{pmatrix} p \\ -q \end{pmatrix}. \quad (3)$$

Write Φ_t for the flow map, the map sending $x(0)$ to $x(t)$. Near any point the flow is governed by the Jacobian of the right-hand side,

$$\nabla(J\nabla H)(x) = J\nabla^2 H(x), \quad \nabla^2 H = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad (4)$$

and $\nabla^2 H$ is symmetric. For a linear system $\dot{x} = Ax$ the flow map is $\Phi_t = e^{At}$.

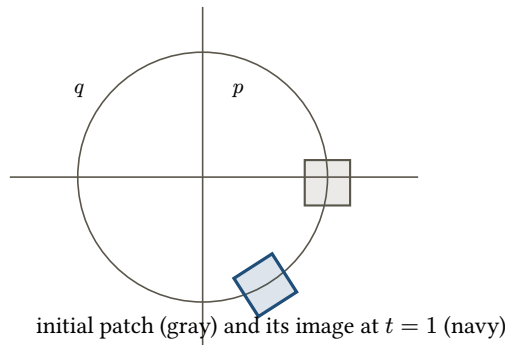


Figure 2: The phase plane of Equation 3 with one small square patch at $t = 0$ and its image under the flow at $t = 1$, both on the orbit through the patch center. Inspect the shape and the area of the image.

Facts to reason with.

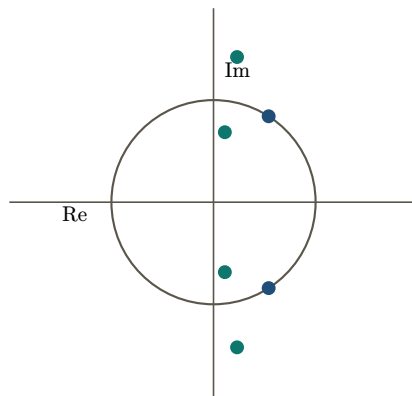
- For Equation 3, $\Phi_1 = \begin{pmatrix} 0.5403 & 0.8415 \\ -0.8415 & 0.5403 \end{pmatrix}$, so $\|\Phi_1^T J \Phi_1 - J\| \approx 1.1 \times 10^{-16}$ and $\det \Phi_1 = 1$.
- Adding damping gives $\dot{q} = p$, $\dot{p} = -q - cp$ with $c > 0$, and $\det \Phi_t = e^{-ct}$; at $c = 0.5$, $\det \Phi_1 = 0.6065$.
- Open [Control/symplectic-matrix/symplectic-matrix-demo.html](#) beside this note and move the stiffness, damping, and integrator controls to watch the same patch and its area ratio.

3. What does symplecticity force on the spectrum?

Question: Take two unrelated symplectic matrices of the same size. If λ is an eigenvalue of one of them, which other complex numbers must also be eigenvalues of that same matrix? Why does that rigidity make it impossible for an asymptotically stable linear system to have a symplectic state-transition matrix?

Facts to reason with.

- For a discrete linear system $x_{k+1} = Mx_k$, asymptotic stability means every eigenvalue of M lies strictly inside the unit circle.
- $M_1 = \Phi_1$ from Equation 3 is symplectic with $\det M_1 = 1$ and eigenvalues $0.5403 \pm 0.8415i$, both of modulus 1.
- $M_2 = e^{H_2}$ for $H_2 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -2 & -0.2 & 0 & -1 \\ -0.5 & 0 & 0 & 2 \\ 0 & 0 & -1 & 0.2 \end{pmatrix}$ is symplectic with $\det M_2 = 1$ and eigenvalues $0.1111 \pm 0.6853i$ and $0.2306 \pm 1.4218i$.
- Both satisfy $\|M_i^T J M_i - J\| < 10^{-15}$.



unit circle; navy M_1 , teal M_2

Figure 3: Eigenvalues of the two symplectic matrices M_1 (navy) and M_2 (teal) in the complex plane, with the unit circle. Inspect the arrangement of the points relative to the circle and to one another.

4. Where does a Hamiltonian matrix enter control?

Question: The LQR problem replaces the plant with a $2n \times 2n$ matrix built from A , B , Q , and R . What structural property does that matrix inherit from the problem, and how does the property deliver the stabilizing solution of the Riccati equation – why exactly n of its eigenvalues are the ones to keep, and what does an invariant subspace have to do with the symmetric solution?

Running example. The same oscillator, now actuated and penalized: $\dot{x} = Ax + Bu$ with

$$A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad Q = \varepsilon I, \quad R = 1, \quad J_c = \int_0^\infty (x^T Q x + u^2) dt. \quad (5)$$

Write $G = BR^{-1}B^T = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$. The optimal control is $u = -Kx$, the classical LQR result – used here, not re-derived – reads $K = B^T P$ off a symmetric solution P of

$$A^T P + PA - PGP + Q = 0, \quad (6)$$

and the controller's solution is the one whose closed loop $A - GP$ is stable. The matrix

$$H = \begin{pmatrix} A & -G \\ -Q & -A^T \end{pmatrix} \quad (7)$$

is called the Hamiltonian matrix of the problem.

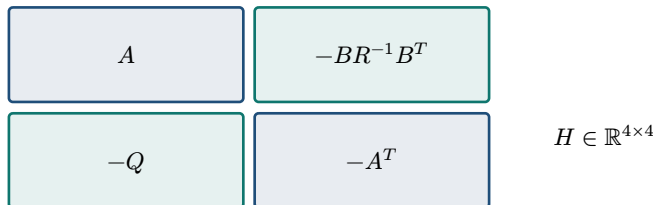


Figure 4: The block layout of Equation 7 for the running example, each block labeled by the problem data it carries. Inspect how the four blocks pair up and which of them is symmetric.

Facts to reason with.

- H satisfies $(JH)^T = JH$ with the 4×4 form $J = \begin{pmatrix} 0 & I_2 \\ -I_2 & 0 \end{pmatrix}$.
- For $\varepsilon = 1$ the characteristic polynomial of Equation 7 is $x^4 + x^2 + 2$, with roots $\pm 0.6761 \pm 0.9783i$: two in the open left half-plane, two in the open right half-plane, none on the imaginary axis.
- If H has no eigenvalues on the imaginary axis, its eigenvectors for the n left-half-plane eigenvalues span an n -dimensional invariant subspace. Writing a basis of it as a stacked block $\begin{pmatrix} X_1 \\ X_2 \end{pmatrix}$ with $X_1, X_2 \in \mathbb{R}^{n \times n}$, the relation $H \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \Lambda$ holds for the stable Λ .

5. What must the numerical reduction preserve?

Question: The stable invariant subspace of H is usually reached through a Schur decomposition of H . Why is an ordinary Schur decomposition of H the wrong instrument for this problem, and what must the transformation preserve so that the computed solution has the symmetry the Riccati equation demands and the stable eigenvalues the controller needs?

REDUCTION	TRANSFORMED FORM	BASIS AND WHAT IT IS BUILT FROM
ordinary Schur	$Q^* H Q = T$, Q unitary, T triangular	columns of Q ; stable and unstable eigenvalues may be interleaved
Hamiltonian Schur	$U^T H U = \begin{pmatrix} T_{11} & T_{12} \\ 0 & -T_{11}^T \end{pmatrix}$, U symplectic-orthogonal	first n columns of U ; the blocks are linked by the form

Table 1: Two reductions of the same Hamiltonian matrix. Inspect what each one guarantees about the transformed matrix and about the basis it hands back.

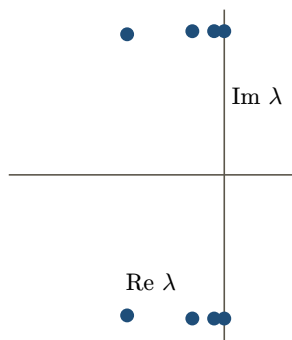
Facts to reason with.

- The Riccati solution must be symmetric, $P = P^T$, and its closed loop must be the stable one; both are properties of the **subspace**, not of any one basis of it.
- The step that reads the Riccati solution off the stable subspace needs its upper block to be invertible and well-conditioned.
- An ordinary Schur decomposition may return the stable eigenvalues in any positions, and its basis is normalized for orthogonality, not for the relation between the two blocks.
- A symplectic-orthogonal U satisfies $U^T J U = J$ exactly, and Hamiltonian Schur reductions are computed by orthogonal symplectic Hessenberg–Schur iterations.

6. What happens when an eigenvalue lands on the imaginary axis?

Question: Weaken the state penalty toward zero. What happens to the Hamiltonian spectrum, to the stabilizing solution of the Riccati equation, and to the invariant-subspace step that was supposed to produce it? What does the position of the spectrum say about whether a stabilizing Riccati solution exists?

Facts to reason with. Keep Equation 5 and Equation 7 but let $Q = \varepsilon I$ shrink. The figure and table below report the left-half-plane member of the closed-loop pair for decreasing ε ; the full spectrum of H is that pair together with its conjugates and their negations.



ε	λ OF $\sigma(H)$ IN THE LHP	$ \lambda $
1	$-0.6761 + 0.9783i$	1.1892
0.1	$-0.2223 + 0.9997i$	1.0241
0.01	$-0.0707 + 1.0000i$	1.0025
0	$0 + 1.0000i$	1.0000

Table 2: The left-half-plane members of $\sigma(H)$ as ε decreases, and the same numbers in the table. Inspect where the points end up and how the real part behaves.

Facts to reason with.

- At $\varepsilon = 0$ the Riccati equation Equation 6 becomes $A^T P + PA - PGP = 0$, which $P = 0$ satisfies, and the closed loop is then A itself, with eigenvalues $\pm i$.

7. Sources and further reading

No external sources have been consulted yet; this note makes no external claims. Every number above is arithmetic on the running example, reproduced by `Control/symplectic-matrix/scripts/verify_symplectic.py`.